Short-Term Rental Prohibitions and Census-Tract Rents

A staggered difference-in-differences design with cohort-aware estimation

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What we ask, and why it is hard

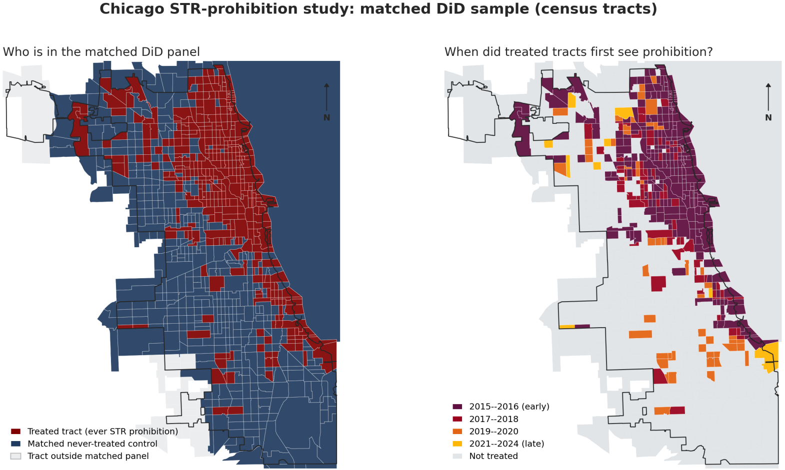
- **Question.** Do census-tract STR prohibitions adopted under Chicago's Shared Housing Ordinance change *rental prices* for long-term tenants?
- **Why hard:** treatment is *staggered* – different tracts adopt in different months – and effects plausibly vary by cohort and time-since-adoption.
- **Implication for design:** pooled two-way fixed effects (TWFE) implicitly compares newly-treated tracts to *already-treated* ones, generating negative weights and biased ATT under heterogeneity (Goodman-Bacon, 2021).
- **Our spine:** the Callaway–Sant'Anna (CS) group-time ATT estimator with *never-treated* controls; pretrend matched sample; ACS + tract linear trend residualization for robustness.

Data

- **Outcome:** ZORI smoothed monthly rent index, ZIP-level, crosswalked to census tract using HUD area shares – a tract-month panel of rents in dollars.
- **Treatment:** first month a tract has *any* parcel covered by an STR prohibition (Chicago Data Portal: prohibited buildings).
- **Sample:** restricted to tracts whose pre-treatment rent slope can be matched to never-treated comparators (k-NN on slope, $k = 3$, ≥ 6 pre-periods).
- **Result:** 373 matched treated tracts, 114 never-treated, 12 cohorts spanning 2016–2019.

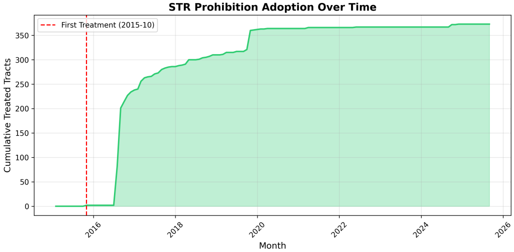
Estimands operate on the rent index in $/month; magnitudes are differences in rent levels, not in growth rates.

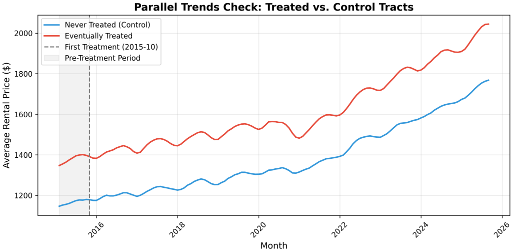
Geography of treatment (matched DiD panel)



Sample: 842 matched tracts — 373 ever-treated, 469 never-treated comparators. Tracts outside the matched panel are shown in light grey.

Adoption dynamics and parallel rents





Rent paths visually parallel pre-prohibition; the level gap motivates residualization.

The level-gap problem (matched sample)

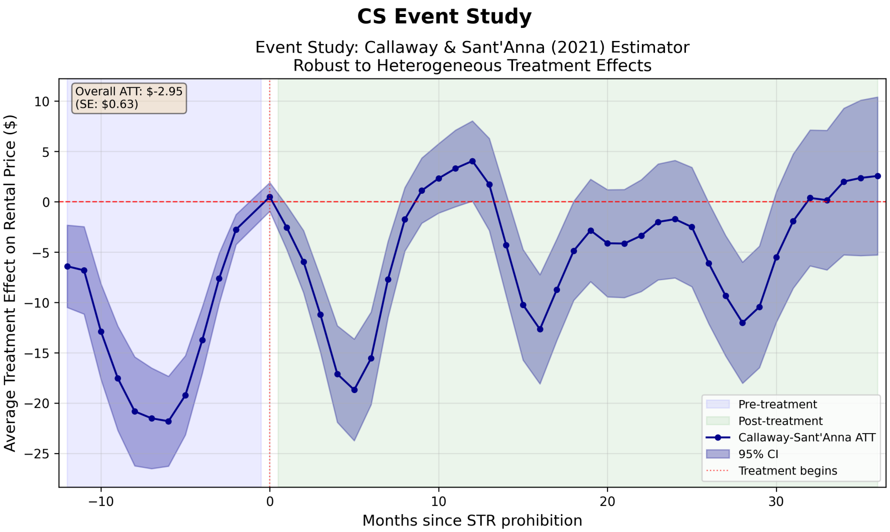
Group	Pre-period mean rent	Std. dev.	n tracts
Ever treated	$1,379.67	313.86	373
Never treated	$1,165.59	252.18	114
Difference (treated − control)	$214.08	Cohen's $d = 0.72$	$p < 10^{-11}$

Source: `pretrend_tract_comparison.csv`. **Implication:** levels differ even where slopes match. Parallel *trends* are the identifying assumption, but the wedge in levels is large enough that any specification that mixes levels and slopes will move the headline number; we report both.

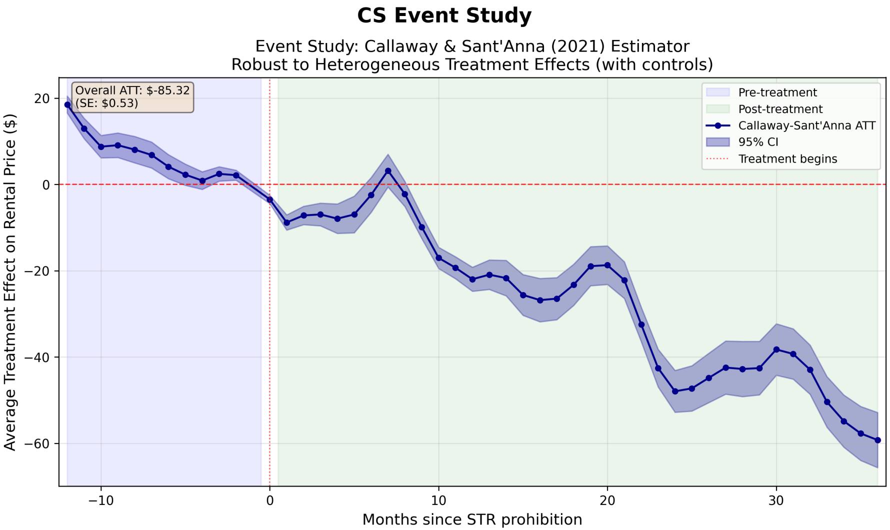
Why Callaway–Sant'Anna (and not pooled TWFE)

- For each cohort g and calendar period t, the CS estimator forms $\widehat{\mathrm{ATT}}(g,t) \;=\; \mathbb{E}[Y_t - Y_{g-1} \mid G_g = 1] \;-\; \mathbb{E}[Y_t - Y_{g-1} \mid C = 1]$, comparing only to *never-treated* (or not-yet-treated) units.
- Aggregation to event-study time e: $\widehat{\mathrm{ATT}}(e) = \sum_g \omega_{g,e}\,\widehat{\mathrm{ATT}}(g, g+e)$ with positive, transparent weights.
- **What this buys us:** no contamination from already-treated controls, no negative weights, and an interpretable per-cohort dynamic effect.
- **Doubly robust analogue (`drc`):** residualize Y on pre-period ACS covariates and tract-specific linear trends, then re-run CS. Aligns with Sant'Anna and Zhao (2020).

Baseline CS event study (no covariate adjustment)



CS event study with ACS + tract-linear-trend controls

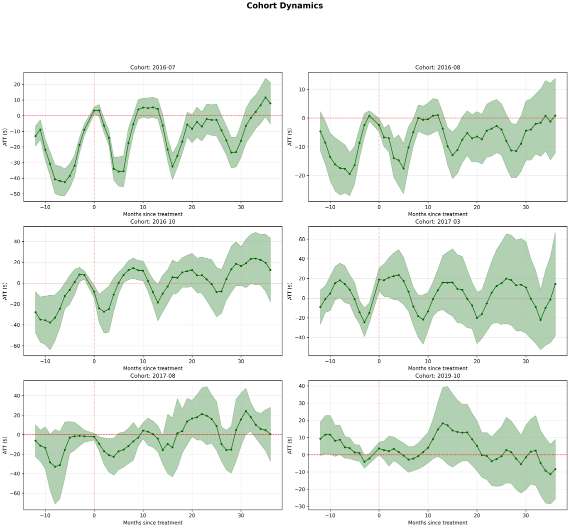


Pooled ATTs: baseline vs. residualized CS

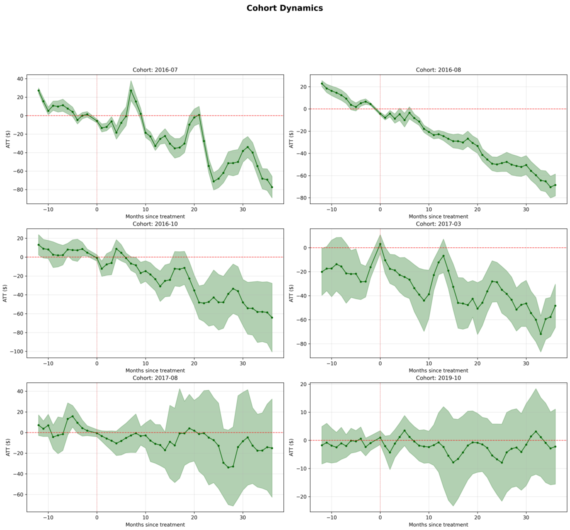
Estimator	ATT ($/mo.)	SE	95% CI
Callaway–Sant'Anna (matched sample)	-2.95	0.63	$[-4.19,\ -1.71]$
CS w/ ACS + tract linear trends (`drc`)	-85.32	0.53	$[-86.37,\ -84.27]$

- Both ATTs are statistically distinguishable from zero; the residualized estimate is interpretable on the residualized scale.
- The gap reflects the level wedge documented earlier: residualization sweeps out tract-specific drift that the unconditional CS leaves in.

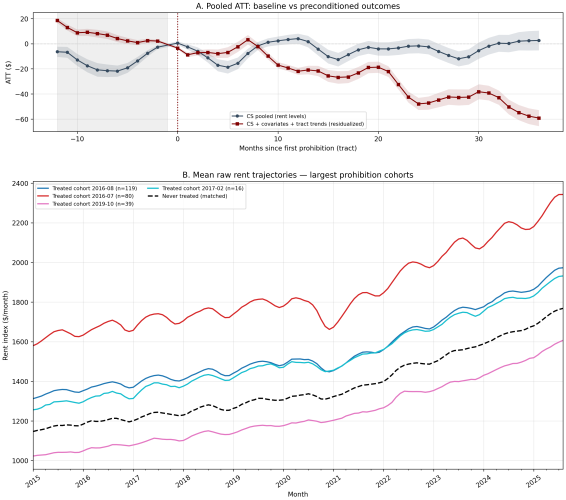
Cohort heterogeneity (baseline CS)



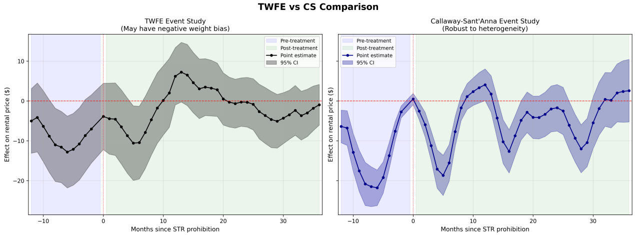
Cohort heterogeneity (CS with controls)



Why residualization moves the headline



TWFE vs. CS: is the bias guarantee binding here?



Why this slide exists

- TWFE uses *already-treated* units as controls for newly-treated ones ("forbidden comparisons") $\Rightarrow$ negative weights, biased ATT under heterogeneity.
- CS only compares to *never-treated* controls – no negative weights by construction.

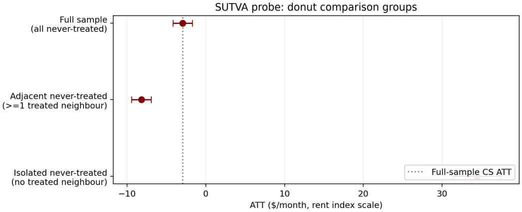
Takeaway

- Trajectories look parallel; mean TWFE$-$CS gap is \$2.83.
- TWFE bias is *not* the binding constraint in this Chicago sample, but we still report CS because its guarantee is unconditional.

Are controls contaminated by neighbour treatment?

- SUTVA failure $\Rightarrow$ never-treated tracts that border prohibition tracts may absorb spillover demand and bias the comparison.
- **Two probes:**
  1. *Donut:* restrict the never-treated pool to tracts with *no* ever-treated neighbour, then to those with at least one.
  2. *Dose:* sort never-treated tracts into quartiles of mid-sample share-of-neighbours-treated; re-estimate pooled CS in each.

Donut comparison: testing for spillover into controls



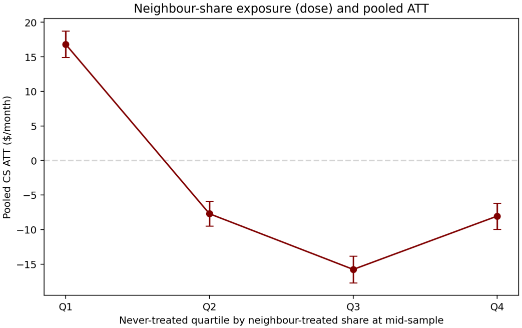
Idea. If never-treated tracts that border treated tracts absorb spillover demand, they look "too treated" and pull ATT toward zero. We split never-treated by neighbour exposure.

- **Full sample:** ATT $= -\$2.95$.
- **Adjacent never-treated** (≥ 1 treated neighbour, $n{=}100$): $-\$8.18$.
- **Isolated never-treated** (no treated neighbour, $n{=}14$): $+\$34.40$.

What we are seeing

- Clean SUTVA failure $\Rightarrow$ *isolated* estimate should be *more* negative; we see the opposite.
- The 14 isolated tracts are peripheral neighbourhoods, structurally different from treated downtown $\Rightarrow$ selection, not spillover.
- Adjacent and full-sample estimates align $\Rightarrow$ controls are not meaningfully contaminated by spillover.

Neighbour-treatment dose



Construction. Sort never-treated controls by mid-sample share of treated neighbours; re-run pooled CS in each quartile.

- Q1 (share 0.13, $n{=}31$): $+\$16.83$.
- Q2 (share 0.44, $n{=}32$): $-\$7.71$.
- Q3 (share 0.66, $n{=}26$): $-\$15.77$.
- Q4 (share 0.92, $n{=}25$): $-\$8.05$.

Why we are seeing this

- Q1 mirrors the donut "isolated" subset: low-exposure controls sit in spatially distinct neighbourhoods.
- Q2–Q4 settle into a stable negative band $\Rightarrow$ no monotone dose gradient; pattern reads as neighbourhood selection rather than dose-dependent spillover.

Pretrend sensitivity (heuristic)

Diagnostic	Value
Max |TWFE coef.| in pre-periods	12.816
Max |second difference| of pre-coefs	3.086
Any strictly positive pre-coef.?	No

- Reports the maximum pre-period TWFE coefficient and the coarsest discrete second-difference (a smoothness budget) over pre-coefs.
- *Spirit only*: a calibrated bound under the Rambachan–Roth (2023) framework would replace these heuristics in a fully "honest" sensitivity exercise.

What we conclude

- **Tract-level STR prohibitions** are associated with a small *negative* change in tract rents on the unadjusted CS scale and a larger negative effect once tract-specific drift and ACS composition are netted out.
- **Design implication:** reporting both the "raw" CS and the residualized CS is honest: residualization changes the meaning of the ATT (it is now on the residual scale), not just its magnitude.
- **Robustness:** cohort heterogeneity is real; spatial probes do not flag obvious SUTVA failure; pretrend coefficients are small in magnitude.
- **Limits:** tract rents are interpolated from ZIP-level ZORI; "treatment" is binary even though prohibition density varies; no calibrated R&R bound is reported here.

Methods references

- Callaway, B. and Sant'Anna, P. (2021). *Difference-in-differences with multiple time periods*. Journal of Econometrics 225(2), 200–230.
- Goodman-Bacon, A. (2021). *Difference-in-differences with variation in treatment timing*. JoE 225(2), 254–277.
- Sant'Anna, P. and Zhao, J. (2020). *Doubly robust difference-in-differences estimators*.
- Rambachan, A. and Roth, J. (2023). *A more credible approach to parallel trends*. RESTUD 90(5), 2551–2598.

- Pipeline source: `src/housing/scripts/did_pipeline_callaway_santanna.py`; documentation in `docs/CS_WITH_CONTROLS.md`.